

PLC Program Scan

Cycle, Continuity, and Execution Patterns

1 The Program Scan Cycle

Inputs to a PLC can change at any time. To process these changes, the controller continually repeats a **scan cycle** whenever it is in **RUN mode**.

Scan Cycle Time

The **scan cycle time** is the total time required to complete one full cycle of the processor's tasks.

- **Reaction Speed:** It determines how fast the scanner reacts to input changes.
- **Typical Duration:** Usually ranges from **1 ms to 10 ms**.
- **Rule of Thumb:** The scan time should be **less than twice** ($< 2\times$) the fastest input change time to ensure no signals are missed.

1.1 Factors Influencing Scan Time

The total scan time is a function of:

- **Processor Speed:** The clock rate of the PLC's internal CPU.
- **Program Complexity:** The total length and logic depth of the ladder program.
- **Instruction Types:** Math-intensive or complex functions take longer than basic bit logic.
- **I/O Density:** The total number of physical input and output points to be scanned.

2 Continuity: Electrical vs. Logical

Understanding how a PLC "solves" a rung requires distinguishing between two types of continuity.

- **Electrical Continuity:** In a hardwired circuit, this is the physical path required for current to flow and energize a load.
- **Logical Continuity:** In a PLC rung, this is a continuous path of "TRUE" (closed) conditional instructions. This is required for the output bit to be set to **1 (ON)**.

Rung Evaluation

The controller evaluates instructions based on the **rung-condition-in**. It then sets the **rung-condition-out** to match the result of that evaluation, passing the "logic flow" to the next instruction.

3 Detailed Scan Process

3.1 Single Rung Operation (Example: Motor Starter)

This example follows the process from pushing a button to a motor turning ON.

Step	Phase	Action	Result
1	External	Push button closes; voltage detected at I:3/6.	Hardware detect
2	Input Scan	Bit I:3/6 is set to 1 in the Input Image Table.	Table update
3	Prog Scan	Evaluate rung: I:3/6 is TRUE; Logical continuity exists.	Solve logic
4	Prog Scan	Set bit O:4/7 to 1 in the Output Image Table.	Table update
5	Output Scan	Write 1 to O:4/7 terminal; Motor starter energizes.	Motor ON
6	External	Button is released; voltage at I:3/6 drops.	Hardware change
7	Next Scan	I:3/6 set to 0 → Rung is FALSE → O:4/7 set to 0.	Motor OFF

Table 1: Single Rung Scan Sequence

3.2 Multiple Rung Logic Flow

PLCs process rungs **sequentially** from top to bottom.

1. **Input Update:** Sense voltage at all terminals and update the entire Input Image Table at once.
2. **Logic Execution:** Using the snapshot in the input table, evaluate each rung in order.
3. **Logic Result:** If conditions are met, write **1** in the output table; otherwise, write **0**.
4. **Output Update:** Transfer the entire Output Image Table to the physical output modules simultaneously.

4 Scan Patterns

There are two primary patterns used by different PLC manufacturers to accomplish the scan function:

1. Horizontal Scan (Rung Method)

The processor solves each rung completely from **left-to-right** before moving to the next rung down. This is the standard pattern for most modern PLCs (like Allen Bradley).

2. Vertical Scan (Column Method)

The processor scans each column of instructions from **top-to-bottom**, moving across the page column by column. This is less common and found in older or specialized controllers.

The scan cycle ensures the digital logic remains in sync with the physical world.